IT4100 – Software Quality Assurance
Year 4 Semester 2

Lab 2 – Code Coverage Analysis

Group Details

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BSc (Hons) in Information Technology
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Task 2

Table 1: Test cases to test the Ward Application.

#Test Case	Test Inputs		Expected Result	Actual Result	Test Result (Pass/Fail)
	Admitted	Discharged			
1	20	5	Surplus = 15 (20-5) ***(25>surplus >=10)*** Move 5 patients to the next ward and Remaining can be added to this ward with more medical facilities	Move 5 patients to the next ward. Admit 10 patients to this ward, Request for more medical officers	Pass
2	62	10	Surplus = 52 (62-10) ***(surplus >= 50)*** Alert that the medical facilities are insufficient.	ATTENTION! Insufficient medical facilities!	Pass
3	2	5	Surplus = -3 (2-5) ***(surplus < 0)*** All patients can be added to the ward without any issues.	All patients admitted without any issues Ward is not in a critical stage	Fail
4	50	20	Surplus = 30 (50-20) ***(50>surplus >=25)*** 5 to this ward, 10 to next ward, remaining to another hospital.	Move 15 patients to the next hospital Move 10 patients to the next ward Admit 5 patients to this ward, Request for more medical officers	Pass
5	40	38	Surplus = 2 (40-38) ***(5>surplus)*** Patients can be added to this ward.	Admit to this ward	Pass

Task 3

Statement Coverage	Statement Coverage = (Executed Statements / Total Statements) * 100% Statement Coverage = (39/40) * 100% = 97.5%
Decision Coverage	Decision Coverage = (Decision outcomes executed / Total decision outcomes) * 100% Decision Coverage = (10/12) * 100% = 83.33%
Path Coverage	Path Coverage = (Independent paths executed / Total paths executed) * 100% Path Coverage = (5/7) * 100% = 71.42%

Task 4

#Test Case	Test Inputs		Expected Result
	Admitted	Discharged	
6	5	20	All patients admitted without any issues AND Ward is not in a critical stage
7	14	8	Patients can be added to this ward with more medical facilities

Task 5

#Test Case	Test Inputs		Expected Result	Actual Result	Test Result (Pass/Fail)
	Admitted	Discharged			
1	20	5	Surplus = 15 (20-5) ***(25>surplus >=10)*** Move 5 patients to the next ward and Remaining can be added to this ward with more medical facilities	Move 5 patients to the next ward. Admit 10 patients to this ward, Request for more medical officers	Pass
2	62	10	Surplus = 52 (62-10) ***(surplus >= 50)*** Alert that the medical facilities are insufficient.	ATTENTION! Insufficient medical facilities!	Pass
3	2	5	Surplus = -3 (2-5) ***(surplus < 0)*** All patients can be added to the ward without any issues.	All patients admitted without any issues Ward is not in a critical stage	Fail
4	50	20	Surplus = 30 (50-20) ***(50>surplus >=25)*** 5 to this ward, 10 to next ward, remaining to another hospital.	Move 15 patients to the next hospital Move 10 patients to the next ward Admit 5 patients to this ward, Request for more medical officers	Pass
5	40	38	Surplus = 2 (40-38) ***(5>surplus)*** Patients can be added to this ward.	Admit to this ward	Pass

6	5	20	Surplus = -15 (5-20) ***(0>surplus<-10)*** Patients can be added to this ward with more medical facilities.	All patients admitted without any issues	Fail
7	14	8	Surplus = 6 (14-8) ***(10>surplus>=5)*** All patients can be added to the ward without any issues, and the ward is not in a critical stage.	Admit to this ward, Request more medical officers and admit to this ward	Pass

Task 6

Code segment with error	Code segment after fixing the error
if(surplus > -10) System.out.println("Ward is not in a critical stage");	if(surplus < -10) System.out.println("Ward is not in a critical stage");